

src/perp_lab — 172 Python files · 42,380 lines · 22 packages

Data plane

data	10 · 1,261	acquisition, causal bars, dev/holdout splits, provenance
validation	4 · 503	schema + quality report, walk-forward CV, purge/embargo
features	7 · 1,738	config-driven causal feature engine
labeling	2 · 483	triple-barrier event labels

Research engine

strategies	19 · 2,592	interpretable baseline families
crt	9 · 4,856	candle-range battery (9 families)
backtesting	3 · 489	vectorised next-open engine with costs + funding
search	11 · 4,213	shared spaces, evaluator, random search + GA
experiments	4 · 1,253	walk-forward + multi-seed orchestration
regimes	3 · 422	fold-fit regime transforms
stochastic	3 · 701	synthetic markets that audit the pipeline
meta_labeling	6 · 2,391	pre-registered supervised ML layer

Evidence plane

evaluation	8 · 2,862	robustness battery C1-C6, multiple testing, DSR/PBO
catalog	8 · 3,249	SQLite registry + DuckDB analytics + artifact store
tracking	4 · 567	run ids, identity fingerprints, environment capture
reporting	11 · 4,754	house style, artifact export, closure reports
eda	27 · 3,853	reusable EDA functions (Ch. 5)

Consumption plane

api	19 · 2,456	read-only FastAPI over frozen artefacts
dashboard	3 · 823	legacy Streamlit inspector (the Next.js panel lives in apps/web)

Cross-cutting

config	4 · 1,693	pydantic contracts for data + experiment YAML
utils	5 · 264	hashing, UTC time, logging, deterministic seeds
cli.py	2 · 957	thin command layer; orchestration stays in modules

Companion trees: tests/ (1,539 pytest), scripts/ (deterministic notebook + figure builders), configs/ (frozen contracts), apps/web (Next.js panel, 112 vitest), docs/decisions (18 ADRs), artifacts/runs (447 evidence packages).